

A · Mechanism Understanding

A1 · Positive mechanism — what drives the advantage

- > **High** Koopman lifting ablation
- > **High** Temporal attention ablation
- > **Med** Feature space interpretability

code access

retraining

now

A2 · Generalisation boundary — why Weather yes, ETTh no

- > **High** Persistent homology TDA
- > **Med** Synthetic data interpolation (ODE → ETTh statistics)

now

B · Improve Generalisation

B1 · Chaotic systems (in-distribution)

- > **Med** Training diversity expansion
- > **Low** Pretraining objective modification
- > **Low** SDE training data
- > **Low** Mean regression mitigation

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B2 · Non-chaotic systems (out-of-distribution)

B2a · Non-chaotic ODEs

- > **Med** Harmonic / VanderPol mechanism investigation
- > **Med** Complexity continuum extension

B2b · PDEs

Chaotic PDEs

- > **Med** Lorenz-96 (inertial manifold hypothesis)
- > **Med** KS equation / Navier-Stokes

Non-chaotic PDEs

- > **Med** Burgers artifact test (PCA vs raw spatial)
- > **Med** Diffusion equations / smooth PDEs

B2c · Real-world time series / sensors

Homogeneous sensors

- > **High** G-SWaN node embeddings (learned per-sensor vector)

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Heterogeneous sensors

- > **Med** XXLTraffic evaluation
- > **Med** Graph topology attention

Specific failing datasets (ETTh)

- > **Med** Adapter / LoRA fine-tuning
- > **Med** Panda-Chronos hybrid adapter

C · Evaluation Standardisation

- > **High** CRPS / Hellinger distance for chaotic systems
- > **Med** Long-horizon distributional evaluation
- > **Med** Cross-regime metric standardisation (DeepLevy)

now

now

Priority: **High** **Med** **Low** Feasibility: **now** **code access** **retraining**